Kokich & Kokich 2006 — likely questions

Congenitally missing mandibular second premolars: clinical options · AJODO 130:437–44

⭐ *= the ten most likely. Page numbers are journal pages (437–444); Proffit chapters are the 7th edition.*

About the cases

⭐ Q. What's the take-home message?

Profile and crowding decide open or close: close if the opposing arch is crowded or the profile protrusive; otherwise keep the space, or close only with extra anchorage. If you keep it, protect the bone: keep a healthy E, extract an ankylosed one early in a growing child without a space maintainer, let moving teeth build the ridge, and place the implant only when serial cephs show growth has stopped. Plan it with the surgeon and restorative dentist.

Source: paper p437, p441–444

Q. What are the key numbers in this paper?

Few: five illustrative cases, no tables. Four to know: Patient 1's E was 13 mm against 7.5 mm for an average lower 5, cut about 2 mm a side toward 7–8 mm; citing Ostler, the ridge loses 25 % in the first 4 years after E extraction and 30 % by 7; citing Fudalej, vertical growth ends on average at about 17 in girls, 21 in boys; and Patient 4's protraction took 4 years.

Source: paper p437, p439–442

Q. Patient 1's E was "submerged". Why didn't they extract it?

Because it wasn't ankylosed: the bone between the E and its neighbours was flat, so it was erupting with them and simply sat a little low. She was 12 and still growing, so the implant had to wait, and keeping the E holds the buccolingual bone. They narrowed it, covered it with composite and closed the remaining space to Class I molar and canine.

Source: paper p437, Fig 1

⭐ Q. How exactly do you reduce the E?

With a sharp carbide fissure or diamond bur. Measure its width at the CEJ on a bitewing, mark it on the occlusal surface, and cut toward the gingiva to take a wafer of enamel and dentin off the mesial and distal, about 2 mm each side, without endangering the pulp. Then bond composite over the exposed dentin, hard to clean once contacts close, and build up the short crown so the opposing tooth can't supererupt.

Source: paper p441–442

Q. Won't the E's divergent roots block space closure?

Usually not: the authors say the roots resorb as the 4 and 6 reach them and bone replaces them, which they call ideal site preparation. Proffit notes the same resorption as a possible problem when the E is meant to stay. Both are right: **good if the E is a placeholder for an implant**, **bad if you want it to last**. The paper doesn't say how long a resorbing E lasts, so I'd monitor it on radiographs.

Source: paper p441–442; Proffit Ch 12

Q. Patient 2: why extract at 8 and then reopen the space? Isn't that double work?

It's deliberate, and the paper admits it adds treatment time; the payoff is a robust ridge without a graft. The Es were ankylosed at 8 with growth ahead, so they came out; with no space maintainer, the 4 and 6 drifted together and their eruption lifted the bone and erased the vertical defect. The lower incisors were too lingual for the chin, so they were proclined and the 5 space reopened, and the separating roots left bone behind: orthodontic implant-site development. Only one of her two sites is shown, with no implant age.

Source: paper p437–438, Fig 2; p443

⭐ Q. Why no space maintainer after extracting the E?

Because a space held open loses bone, while teeth that drift and erupt together carry bone with them. The authors skip them in most cases, especially when an implant is planned, and later push the 4 and 6 apart to rebuild the ridge, at the cost of longer treatment. Proffit agrees in principle: no space maintenance when the 5 is missing and the arch length is meant to be lost, and the drifting first molar brings bone with it.

Source: paper p443; Proffit Ch 11, Ch 12

Q. What was the logic of using an implant as anchorage in Patient 3?

She was an adult missing the lower right 5 and 6: too big a gap for one tooth, too small for two. One implant, placed from a wax-up at the second-premolar site, got a provisional crown and a bracket and served as a fixed anchor to bring the 7 forward into Class I without moving the incisors. **One restoration instead of two**; the cost: an unopposed upper 7, to extract or oppose with an implant if it supererupts.

Source: paper p438–439, Fig 3; p443

⭐ Q. Why a chincup in Patient 4, and what would you use today?

To bring the molars forward without retracting the incisors. Her profile was ideal, so extracting the remaining upper 5 and closing conventionally would have retracted the lips too far relative to the chin. Mini-implants weren't in use yet, so a chincup and elastics slid the 6s forward on a continuous wire: 4 years, Class I molars, no restorations. **Today: miniscrews**, as the paper itself says; Proffit calls bone screws the easiest way to bring posterior teeth forward, from about age 12.

Source: paper p439–440, p443; Proffit Ch 16, Ch 12

Q. In Patient 5, why move the first premolar instead of grafting? Was her E extracted too late?

Vertical grafting is unpredictable and adds cost, and a moving root brings bone with it. The 4 was pushed distally through the collapsed ridge; the site it left took the implant at 17, no graft. The cited support is indirect (Hom & Turley: 14 adults, ridge wider but most lost about 1.3 mm of crestal bone); Eliasova 2014 studied exactly this in 71 sites: width +28.5 %, height +1.1 mm, largely stable 2–5 years later; adults show some buccal dehiscence (Ramos 2020). On timing, the paper cites her in its paragraph on extracting too late (take an ankylosed E out before a vertical defect forms), but doesn't say when her extraction was decided, so I wouldn't call it a mistake.

Source: paper p440, p442–443 (refs 10–13); Hom & Turley 1984 (abstract); Eliasova 2014; Ramos 2020

About the evidence

⭐ Q. What level of evidence is this paper?

Low: a Clinician's Corner, expert opinion illustrated by five selected cases: level 5 on the Oxford CEBM scale, or level 4 at best if you count the cases as a case series. No methods, sample, measured outcomes, follow-up or failures; 5 of the 22 references include the first author (23 %, derived), natural when a Clinician's Corner distils the authors' own work. Nor has any trial compared closure with implants: Thilander's 2008 review found only 11 relevant papers, mostly retrospective or case reports. Its value is the framework.

Source: paper p437–444; Thilander 2008 (J Oral Rehabil 35 Suppl 1:64–71)

Q. What are its main strengths and weaknesses?

Strengths: a clear framework (profile, crowding and incisor position decide; growth is the clock), practical techniques, five strategies each shown as a case (hemisection, facemask and miniscrews are only described). Weaknesses: **selection bias**, no outcome numbers, and key claims unreferenced (reduction, no space maintainer, the crowding and profile rule) or backed indirectly: an animal study for loading, a case report for closure, grafting studies at other sites for unpredictable vertical grafts.

Source: paper p441–444; refs 10, 11, 18, 19

⭐ Q. Why an implant and not a bridge?

In a young patient a conventional bridge risks the abutments' pulps, and posterior resin-bonded bridges have a weaker record: in Creugers 1998, agenesis (CRR 2.9) and the mandible (CRR 2.2) were both risk factors for failure, exactly this patient. Single implants survive 97.2 % at 5 years and 95.2 % at 10 (Jung 2012). Resin-bonded bridges now do better than the older data the paper cites (91.4 % at 5 years, 82.9 % at 10; Thoma 2017), so they're a fair provisional while growth finishes.

Source: paper p441 (refs 2–6); Creugers 1998; Jung 2012; Thoma 2017; Proffit Ch 19–20

Q. How much does the ridge resorb after you extract the E, and why does it matter?

About a quarter, mostly early. The paper quotes Ostler and Kokich: 25 % in the first 4 years and 5 % more by 7, yet still wide enough for an implant; the original abstract (35 sites, 22 patients) says 25 % within 3 years, then 4 % over the next 3. More is lost facially, so the implant sits lingual and cusp loading must be adjusted, or the abutment or crown may fracture.

Source: paper p441 (refs 7, 8); Ostler & Kokich 1994 (abstract; facial-side detail not in it)

Q. The paper says orthodontic load stimulates bone around implants. Is that proven?

Not in patients: the cited study is in rabbits. Duyck 2001 loaded already-integrated implants in tibiae for 14 days: static load matched unloaded controls histologically, with no bone loss; dynamic load caused crater defects. So **static load did no harm, but no extra bone was shown**. In humans, the 7 implants retrieved after 2–12 months of orthodontic load (41 adults treated) were all osseointegrated (Trisi & Rebaudi 2002).

Source: paper p443 (ref 18); Duyck 2001; Trisi & Rebaudi 2002

🩺 What would you do if...

⭐ Q. You see a submerged E with no successor. How do you tell if it's ankylosed, and what do you do?

Look at the bone between the E and the 6 and 4: flat means they're erupting together; oblique bone, lower around the E, confirms ankylosis (Fig 1B vs 2B), a sign from the authors' own review (ref 1), not an accuracy study. Clinically: infraocclusion, a dull percussion note and no mobility, though percussion is subjective and changes only once about a fifth of the root is fused. Then decide by growth: **a lot left → extract early**, before a vertical defect needs an unpredictable graft; **little left and slight infraocclusion → keep it**. Proffit adds decoronation for the growing child (remove the crown, bury the roots, keep the ridge); the paper doesn't discuss it, but I'd consider it before an implant.

Source: paper p442–443 (ref 1); Arhakis & Boutiou 2016; Proffit Ch 12

⭐ Q. When would you simply close the space?

When the opposing arch is crowded or the profile is protrusive. With no crowding and a normal profile, closure could flatten the lips, so keep the space or close it with extra anchorage: facemask or chincup outside, miniscrews inside. Proffit adds that one missing premolar usually means a restoration, and that unopposed 7s can over-erupt.

Source: paper p443; Proffit Ch 12

⭐ Q. If you close a lower 5 space and leave the upper arch alone, what molar relationship do you finish in?

Toward Class III, up to a full unit, since the lower 6 comes forward by up to a premolar width (derived). Unless 5s are missing in all four quadrants, Proffit says opposing-arch extractions may be needed for a near-ideal Class I, and an unopposed upper 7 can over-erupt. That's why Patient 4's spaces were closed in both arches.

Source: Proffit Ch 12; paper p439, p443 (derived)

Q. A 9-year-old with a missing lower 5: what do you do?

First confirm it on a repeat radiograph: late-forming premolars occur. If closure suits the profile and crowding, early extraction or hemisection of the E lets the 6 drift forward (Proffit's window is 7–9 years, though drift is unpredictable; Mamopoulou: 80 % of the space closed within 4 years of extraction at a mean 11.8 years, partly by tipping). If the space will be kept, a healthy E stays; an ankylosed one with growth ahead comes out, with no space maintainer.

Source: Aasheim & Ogaard 1993; Proffit Ch 12; Mamopoulou 1996; paper p443

Q. And a 12-year-old with a healthy E?

That's Patient 1: keep the E, narrow it and cover it with composite, and let it hold the bone until serial cephs show growth has stopped (about 17 in girls, 21 in boys). If it's ankylosed and still submerging with growth ahead, extract it now. TADs can be used from about 12.

Source: paper p437, p441–443; Proffit Ch 12

Q. How much space do you open, and how do you hold it until the implant?

About a premolar's width (7.5 mm on average), set on a wax-up with the restorative dentist, with the roots of the 4 and 6 checked parallel on a radiograph (standard practice, not in the paper). Then a fixed retainer, almost always best per Proffit; the paper doesn't cover retention.

Source: paper p437, p443; Proffit Ch 19–20

Q. A 16-year-old? What's the prognosis if the E is simply kept?

Good if it's healthy: Bjerklin and Bennett followed 59 Es to a mean of 20.5 years and lost 7, none after 20; worse with marked infraocclusion, caries or restorations, or over half the root resorbed (Patano 2023). If it's lost with a narrow ridge, move the 4 as in Patient 5 or graft, and place the implant once serial cephs show growth has stopped. Closing instead without retracting the incisors means miniscrew protraction: about 0.55 mm a month, roughly 2.7 years of treatment in one small cohort.

Source: paper p440, p443; Bjerklin & Bennett 2000; Patano 2023; Taneja 2024

Q. An adult?

A healthy retained E stays: in Sletten's 28 adult Es followed about 12 years, 86 % still worked, with 0.16 mm of root shortening. If it's lost: an implant (lingual position, maybe a graft), or closure, conventional if the opposing arch is crowded or the profile protrusive, on TADs if the profile must not change, accepting slow movement (old narrowed sites take about 3 years, Proffit) and possible bone loss distal to the 7. If the 6 is missing too, use the Patient 3 approach.

Source: Sletten 2003; Baik 2019; paper p443; Proffit Ch 20

📚 Beyond the paper (literature & Proffit)

⭐ Q. Why wait until about 17 in girls and 21 in boys, and how do you know growth has stopped?

An implant behaves like an ankylosed tooth: placed while the face is still growing vertically, it is left behind as the neighbours erupt. So the paper times it on superimposed serial cephs, not age (Patient 5's implant waited for them); Proffit calls serial cephs the most accurate test, and hand-wrist and vertebral staging inaccurate for this. 17 and 21 are averages the paper cites from Fudalej, whose published abstract gives no cut-off ages. **It's not a guarantee**: Bernard 2004 found steps up to 1.86 mm beside anterior maxillary implants even in adults aged 40–55, and long- or short-face patients stay at risk after 20 (Heij 2006), so I warn patients the crown may need remaking.

Source: paper p440–441; Fudalej 2007 (abstract); Bernard 2004; Heij 2006; Proffit Ch 6, Ch 20

Q. How common is this?

About 6 % of people miss a tooth other than third molars (6.4 % in Khalaf's 2014 review; Polder 2004: 4.6 % of European males, 6.3 % of females). In Caucasians the lower 5 is the commonest (about 3 %; 3.4 % in a 2026 Spanish sample), ahead of the upper lateral and upper 5; rankings vary by population.

Source: Polder 2004; Khalaf 2014; Garcia-Navas 2026; Schonberger 2023

Q. When can you be sure a second premolar is missing?

Not before about 8–9, and not completely even then: Proffit says good premolars seldom form after 8, yet Aasheim and Ogaard found that 11.3 % of boys and 2.9 % of girls "missing" a 5 at 9 formed one by 12. So repeat the radiograph before anything irreversible; the paper doesn't say how it was confirmed in 8-year-old Patient 2.

Source: Proffit Ch 12; Aasheim & Ogaard 1993

Q. What about hemisection?

The paper favours it for a child seen early: section the E so the 6 erupts mesially without moving the incisors (Northway 2004); Proffit adds pulp therapy for the mesial half. But a 2025 split-mouth RCT in 40 children found no clinically relevant gain over plain extraction (residual space 2.04 vs 2.39 mm, p = 0.053), with more complications and cost, so **it's an option, not the standard**.

Source: paper p443–444; Northway 2004; Proffit Ch 12, Ch 16; Abdul Jabbar 2025

Q. Why isn't autotransplantation in the paper, and would you consider it?

It isn't discussed, though the abstract of the paper's own reference 19 mentions it. **In a growing patient it's attractive** because a transplant keeps the alveolus developing, which an implant can't: Josefsson 1999 reported 92 % success in lower-5 agenesis sites with incomplete donor roots, and Czochrowska 2002 found 90 % survival (79 % success) at a mean of 26 years. Donor: an upper premolar or a third molar.

Source: Fines 2003 (ref 19, abstract); Josefsson 1999; Czochrowska 2002

Q. If you protract molars on miniscrews, what problems do you expect?

Slow movement and long treatment: about 0.55 mm a month and 2.7 years in a small 2024 agenesis cohort (12 TAD patients); in adults, about 32 months with 0.8 mm of root shortening and 0.56 mm of bone-height loss (Kim 2015). Add molar tipping and rotation, miniscrew failure in about 13.5 %, midline shift in unilateral cases (seen even after spontaneous closure, Mamopoulou), and more third-molar eruption.

Source: Taneja 2024; Kim 2015; Papageorgiou 2012; Baik 2016; Mamopoulou 1996

Q. What does Proffit say, and where does he differ?

Chapter 12 reads like a short version of this paper: keep and narrow the E, extract an ankylosed one early without a space maintainer, let the molar bring bone, hemisect, close when crowded or protrusive, and wait for serial cephs before the implant. He adds decoronation and pulp therapy with hemisection, and warns that drift is unpredictable. Two real differences: he flags root resorption of a reduced E as a possible problem when the E is meant to stay, and he treats the chincup as a Class III appliance.

Source: Proffit Ch 6, Ch 12, Ch 13, Ch 20

⚠️ Tricky / trap questions

Q. Did you notice anything odd in the paper?

Only small text–caption slips; none changes the message. The main one: in Patient 3 the text and photos put the implant at the second-premolar site; the caption says first premolar. More substantive: two cited sources say a little less than the text, on ridge loss and loading.

Source: paper p438–442 (text vs legends of Fig 3–5), p441, p443; Ostler & Kokich 1994 (abstract); Duyck 2001 (abstract)

Q. Do the reduction numbers add up?

Not exactly: 2 mm a side from a 13-mm E leaves 9 mm, not 7–8; reaching 7.5 mm needs about 2.75 mm a side (derived). The bitewing CEJ width is the real guide. And 13 mm is a big E: Proffit puts the average lower E about 2 mm wider than the 5; Bjerklin's mean was 10.53 mm.

Source: paper p437, p442 (derived); Proffit Ch 3; Bjerklin & Bennett 2000

Q. Isn't the chincup a Class III appliance? And how did a 5-mm overjet close without retracting the incisors?

Yes: Proffit describes it as mainly rotating the mandible down and back, unwanted in a Class II. Here it was only anchorage for elastics pulling the molars forward. The superimposition shows only that the lower incisors didn't move back (Fig 4F); the paper is silent on the uppers, so some upper retraction is the likeliest way the overjet was reduced (my inference). With no overjet mechanics or vertical change reported, I'd ask to see the upper incisor and mandibular plane.

Source: paper p439–440, Fig 4F; Proffit Ch 13

Q. Did Patient 3 really show "implant before orthodontics" and early loading?

Not quite. The Discussion says implant before orthodontics, but the case aligned first, then wax-up and implant, then closure. The provisional went on after integration (conventional loading); that fits the paper's own rule (load once the provisional is on), and the paper never presents the case as early loading.

Source: paper p438, p443